STATS 402

Comparison of Spatio-Temporal Models (GPR/GAM) vs Traditional ML (LR/RF/SVR) for Nitrogen Fixation Prediction

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Background & Motivation

Research Problem:

- Marine nitrogen (N_2) fixation, the process by which diazotrophs convert atmospheric nitrogen into bioavailable forms, is critical for ocean productivity and carbon sequestration.
- However, global estimates of N_2 fixation remain highly uncertain, and its environmental controls are poorly understood.
- Traditional ML (LR/RF/SVR) in Tang et al. (2019) ignores explicit spatio-temporal dependencies and interactions between space, time, and environmental variables.

Background & Motivation

Why important?

- Spatio-temporal models (GPR/GAM) can capture complex interactions (e.g., time-space autocorrelation, SST-PAR synergy) and uncertainties, improving global estimates.

- Potential applications: Climate modeling, marine nutrient cycling.

Related Work

Tang et al. (2019)
Luo et al. (2014)

- Used LR/RF/SVR with environmental features (PAR, SST, etc.)
- Achieved reasonable accuracy but limited by: (1) No Fe data; (2) No explicit spatio-temporal kernels; (3) Assumed static relationships across regions/seasons.

My Project

- Add dissolved iron (Fe) as a predictor and use 2023 World Ocean Database to address gaps
- Introduce **GPR** (spatio-temporal kernels + uncertainty quantification) and **GAM** (additive interactions + interpretability).

Research Methods

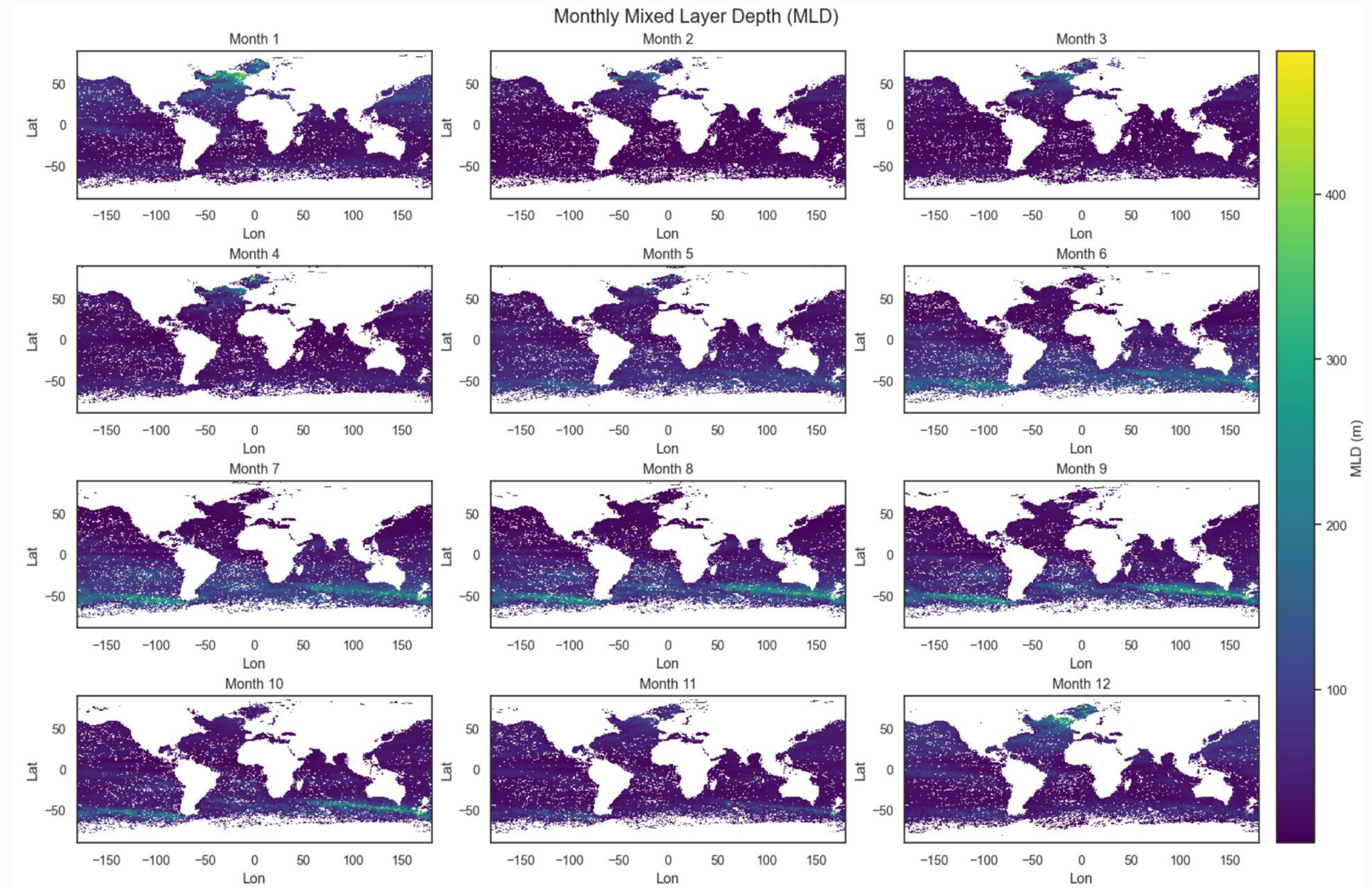
- Data

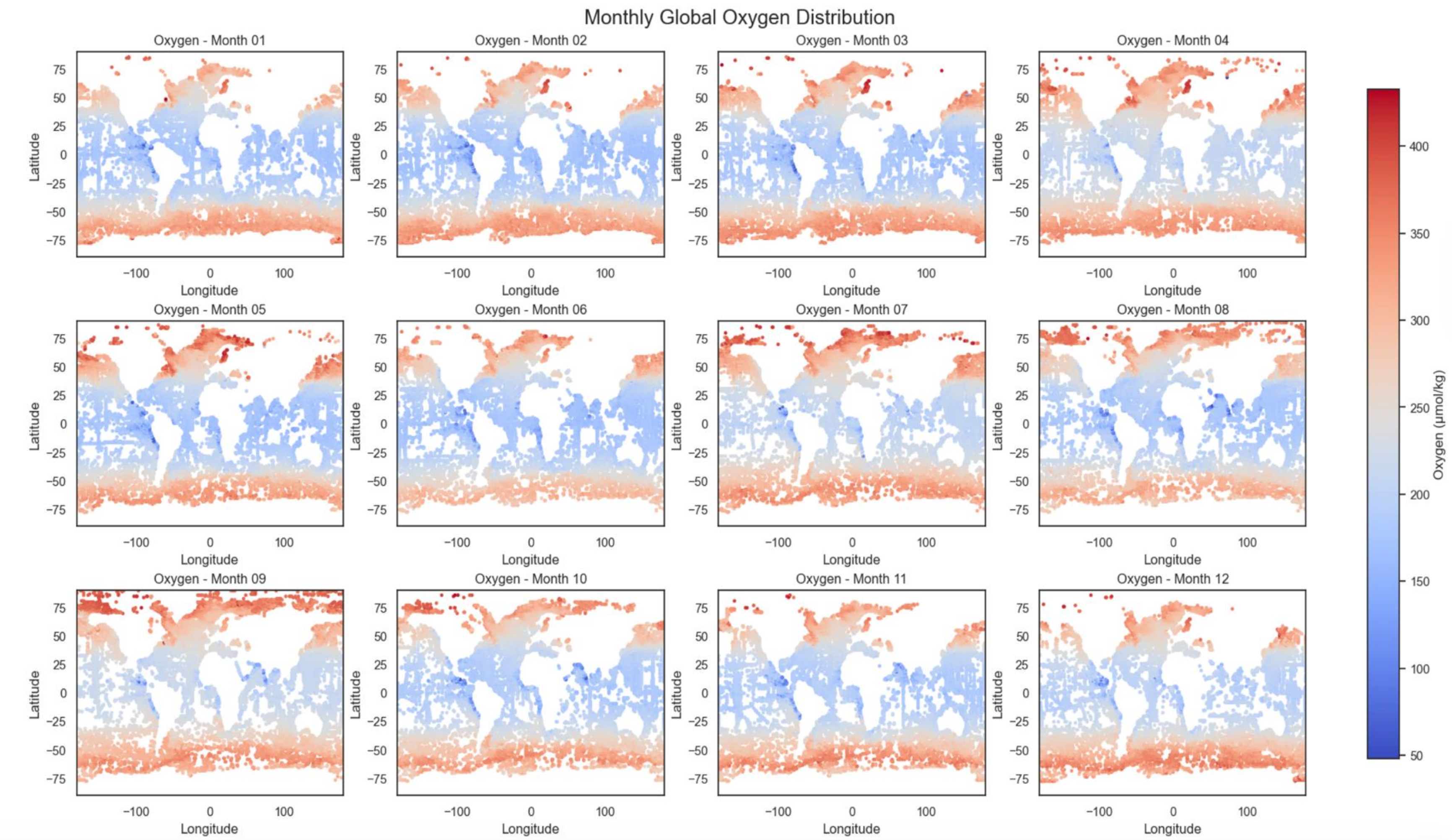
Data Acquisition:

- **Diazotrophs Database:** N2_fixation(24h) [Shao et al., 2023].

- **Satellite Data: Chlorophyll-a (CHL), Photosynthetically Available Radiation (PAR), and Sea Surface Temperature (SST)** [the Oregon State University website, covering measurements at different depths and times (monthly, seasonal, and annual)], **Mixed Layer Depth (MLD)** [the University of California, San Diego's mixed layer database**].**

- **Oceanographic Variables:** Data for **salinity, nitrogen (N), phosphate(P), iron(Fe), and dissolved oxygen (DO)** [the NOAA's World Ocean Atlas 2023, providing 12 months of measurements]





Research Methods - Data

Data Matching:

- **Conversion of Dates:** Observational data dates were converted into 8-day units to align with satellite data time units.

- **Data Integration:** For each unique date (in 8-day intervals), matched values were assigned to the corresponding observational data points. [SST/CHL/PAR]

- **Spatial Matching:** Latitude and longitude coordinates from observational data were matched with corresponding satellite data grid points.

	LATITUDE	LONGITUDE	Total N2 Fixation (µmol N m-2 d-1)	DATE	YRDY	MLD	CHL	SST	PAR	Salinity	...	coord1	coord2	coord3
0	0.000997	-154.999900	23.092953	2006-08-21	2006233	39.282626	0.207077	27.741251	54.784996	35.240	...	0.000017	-0.422620	0.906307
1	22.750000	-158.000000	75.187035	2005-08-14	2005225	46.160884	0.055314	26.434376	53.250000	34.988	...	0.386711	-0.345463	0.855050
2	21.000000	-159.000000	188.927822	2003-08-15	2003225	39.243011	0.054018	26.883751	57.124001	35.042	...	0.358368	-0.334565	0.871572
3	38.464000	-73.300000	531.907504	2009-08-19	2009225	10.202625	0.220413	26.231251	51.389999	31.925	...	0.622023	-0.749974	-0.225003
4	37.855000	-38.642000	132.559691	2009-08-20	2009225	25.805195	0.052617	25.435045	48.073997	36.478	...	0.613665	-0.493047	-0.616701
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

Research Methods - Data

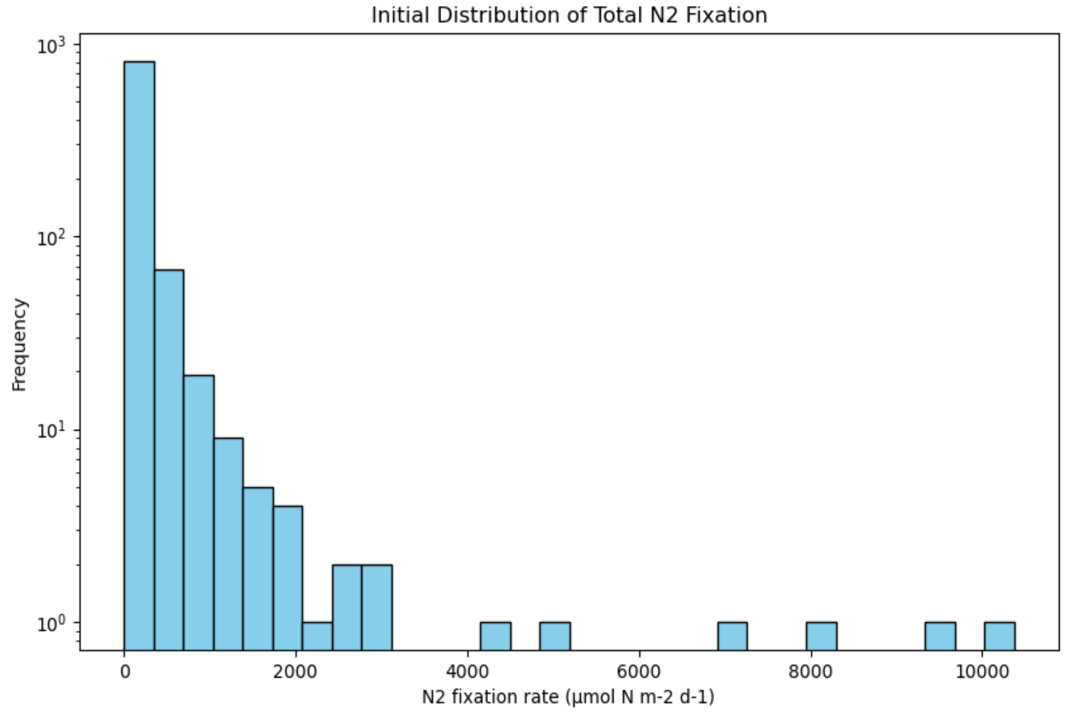
Data	Symbol	Source	Spatial resolution	Temporal resolution	Range	Log10-transformed	p-value
Observed N2 fixation ($\mu mol\ N \cdot m^{-2} \cdot day^{-1}$)	NF	Tang et al. (2019)	Point	Daily	0.0001-5111.6352	Yes	\
Average phosphate within euphotic depth (μM)	P_{eup}	World Ocean Atlas 2013	1°	Monthly climatology	0.0075-2.5801	Yes	≤ 0.01
Sea surface salinity (psu)	SSS		1°		27.535-37.308	No	≤ 0.01
Average nitrate within euphotic depth (μM)	N_{eup}		1°		0.0045-28.4421	Yes	≤ 0.01
Minimum dissolved oxygen in 0–500m (ml/L)	DOmin		1°		0.478-307.398	No	≤ 0.01
Average dissolved iron within euphotic depth (nmol L-1)	Fe_{eup}	Huang et al. (2022)	\	Monthly climatology	0.0768-2.7058	No	≤ 0.05
Sea surface temperature (°C)	SST	SeaWiFS and MODIS	0.083°	8 days	-1.535-31.620	No	≤ 0.01
Chlorophyll-a (mg/m3)	[Chl]		0.083°		0.0107-8.9683	Yes	≤ 0.01
Photosynthetically available radiation ($Einstein \cdot m^{-2} \cdot day^{-1}$)	PAR		0.083°		9.8810-65.5450	No	> 0.05
Mixed Layer Depth	MLD	Ifremer	2°	Monthly climatology	10.2-101.7	No	≤ 0.05
Coord1	COORD1	Tang et al. (2019)			-0.913-0.8661	No	≤ 0.01
Coord2	COORD2				-0.9974-0.9994	No	≤ 0.01
Coord3	COORD3				-0.9999-0.9999	No	≤ 0.01
Time_cos	Time_cos				-1.0-1.0	No	≤ 0.01
Time_sin	Time_sin				-1.0-1.0	No	≤ 0.01

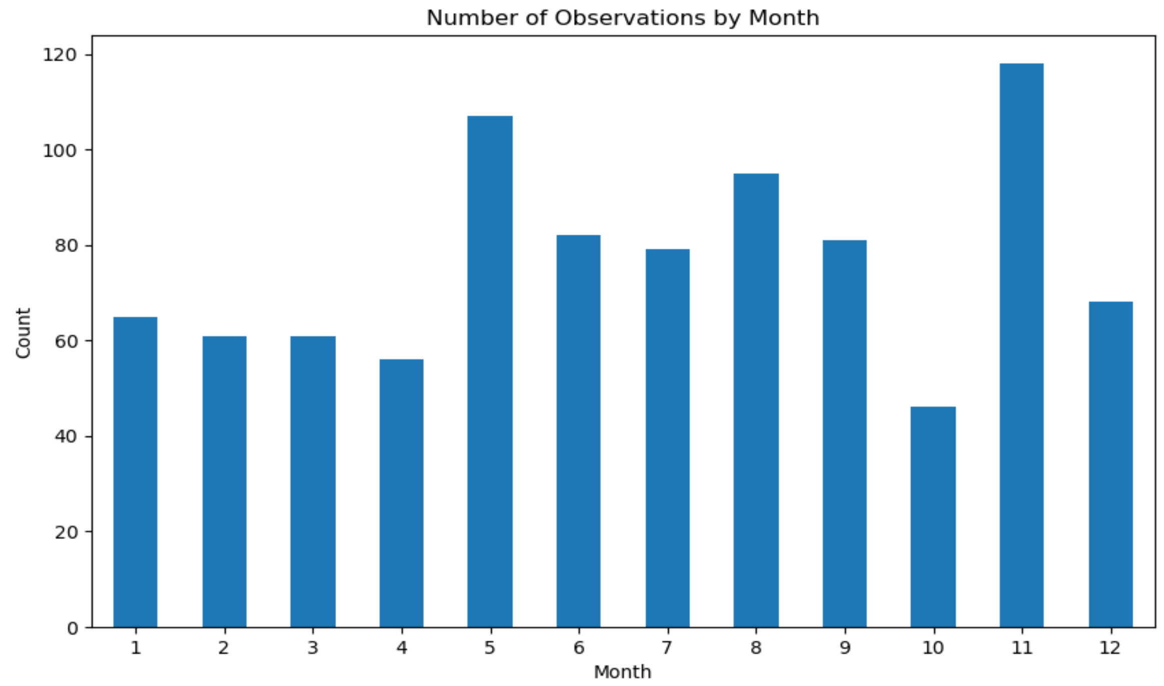
Table 1. N2 fixation and environmental predictors used by the models

close (i.e., January is after December). This is also true for geographical distances in coordinate space (e.g., coordinates −179° and 180°). To address this issue, we transformed the latitude, longitude, and month properties to periodic functions simulated with sine and cosine functions as follows (Gade, 2010; Gregor et al., 2017):

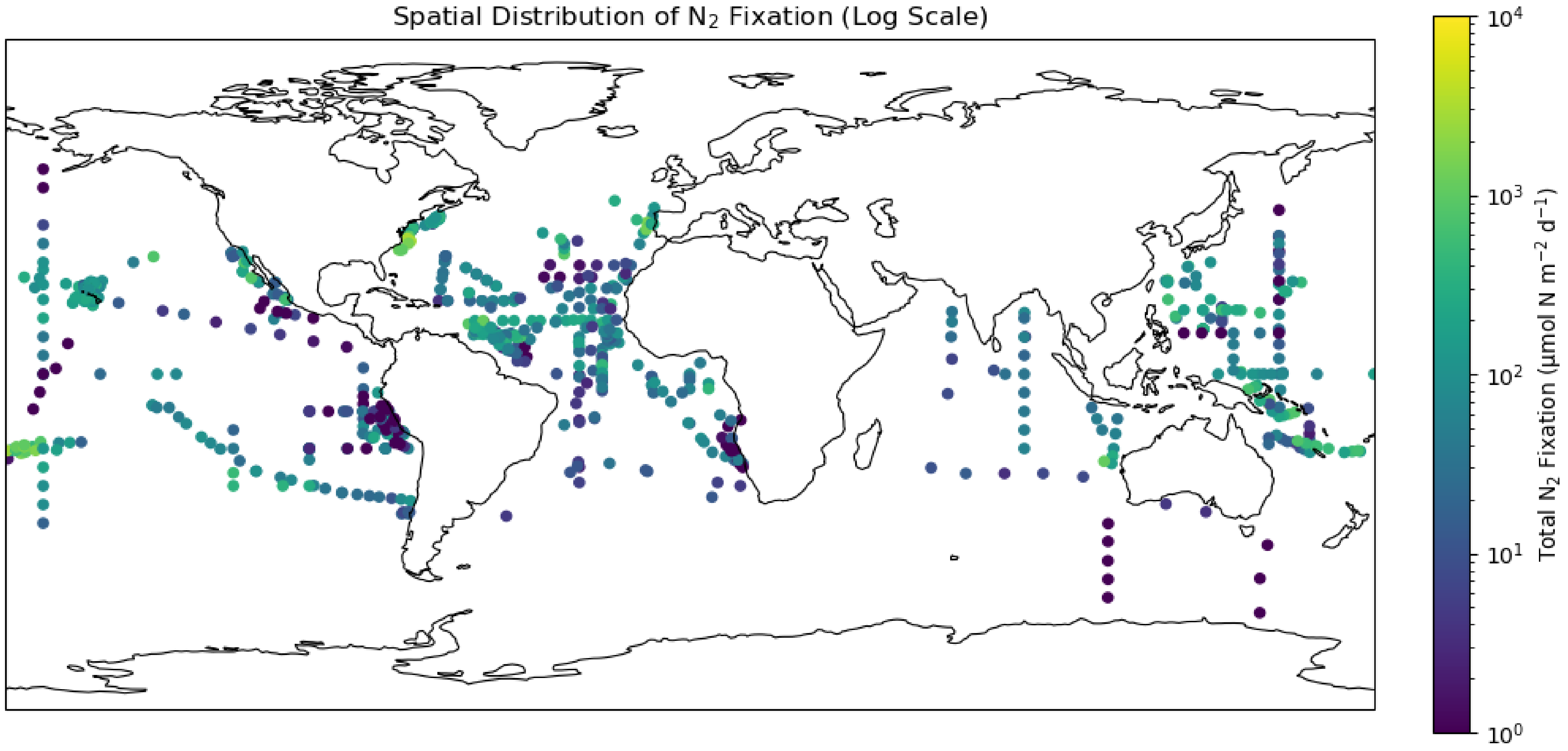
$$\text{coordinates} = \begin{pmatrix} \sin\left(\text{latitude}\cdot\frac{\pi}{180}\right) \\ \sin\left(\text{longitude}\cdot\frac{\pi}{180}\right)\cdot\cos\left(\text{latitude}\cdot\frac{\pi}{180}\right) \\ -\cos\left(\text{longitude}\cdot\frac{\pi}{180}\right)\cdot\cos\left(\text{latitude}\cdot\frac{\pi}{180}\right) \end{pmatrix} \quad (1)$$

$$\text{time} = \begin{pmatrix} \cos\left(\text{month}\cdot\frac{2\pi}{12}\right) \\ \sin\left(\text{month}\cdot\frac{2\pi}{12}\right) \end{pmatrix} \quad (2)$$





Research Methods - Data



- **Log-transformed N_2 fixation rates to address skewed distributions.**
- **Transformations for latitude, longitude, and month to preserve continuity.**
- **80% training set (n=677) and 20% test set (n=169), split based on month.**

Research Methods - Models

Baseline (Tang et al.): LR/RF/SVR

- LR + Ridge Regularization: Ridge (alpha=100)
- RF: n_estimators=300, max_depth=10, max_features='log2'.
- SVR (RBF kernel): C=1, epsilon=0.3, gamma=0.1.
- Optimized with GridSearchCV (hyperparameters, 3-fold cross-validation).

Gaussian Process Regression(GPR): $f(\mathbf{x}) \sim \mathcal{GP}(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x}'))$

- Matérn: spatial coherence, ExpSineSquared: modeling time seasonality (periodicity), RationalQuadratic: environment variables, WhiteKernel: modeling noise.
- Optimized via GridSearchCV; visualized monthly/annual predictions + uncertainty.

Research Methods - Models

Generalized Additive Model (GAM):

$$\hat{y} = \beta_0 + \sum_j f_j(x_j) + \sum_{(i,k)} f_{ik}(x_i, x_k)$$

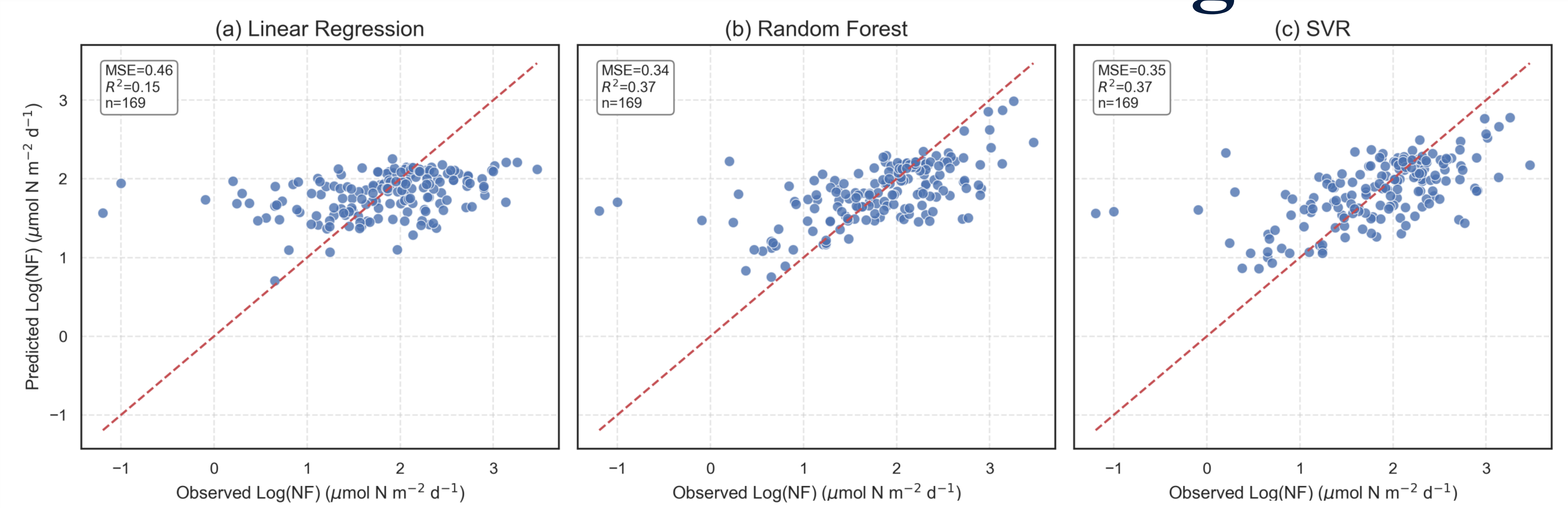
$$\begin{aligned}\hat{y} = \beta_0 &+ f_{\mathrm{PAR}}(\mathrm{PAR}) + f_{\mathrm{SST}}(\mathrm{SST}) + f_{\mathrm{PAR,SST}}(\mathrm{PAR}, \mathrm{SST}) \\ &+ f_{\mathrm{log\_CHL}}(\mathrm{log\_CHL}) + f_{\mathrm{log\_N}}(\mathrm{log\_N}) + f_{\mathrm{log\_CHL,log\_N}}(\mathrm{log\_CHL}, \mathrm{log\_N}) \\ &+ f_{\mathrm{log\_CHL,log\_P}}(\mathrm{log\_CHL}, \mathrm{log\_P}) + f_{\mathrm{log\_CHL,Fe\_avg}}(\mathrm{log\_CHL}, \mathrm{Fe\_avg}) \\ &+ f_{\mathrm{MLD}}(\mathrm{MLD}) + f_{\mathrm{Salinity}}(\mathrm{Salinity}) + f_{\mathrm{DO}}(\mathrm{DO}) \\ &+ f_{\mathrm{coord1,coord2}}(\mathrm{coord1}, \mathrm{coord2}) + f_{\mathrm{coord3}}(\mathrm{coord3}) \\ &+ f_{\mathrm{time\_sin,time\_cos}}(\mathrm{time\_sin}, \mathrm{time\_cos})\end{aligned}$$

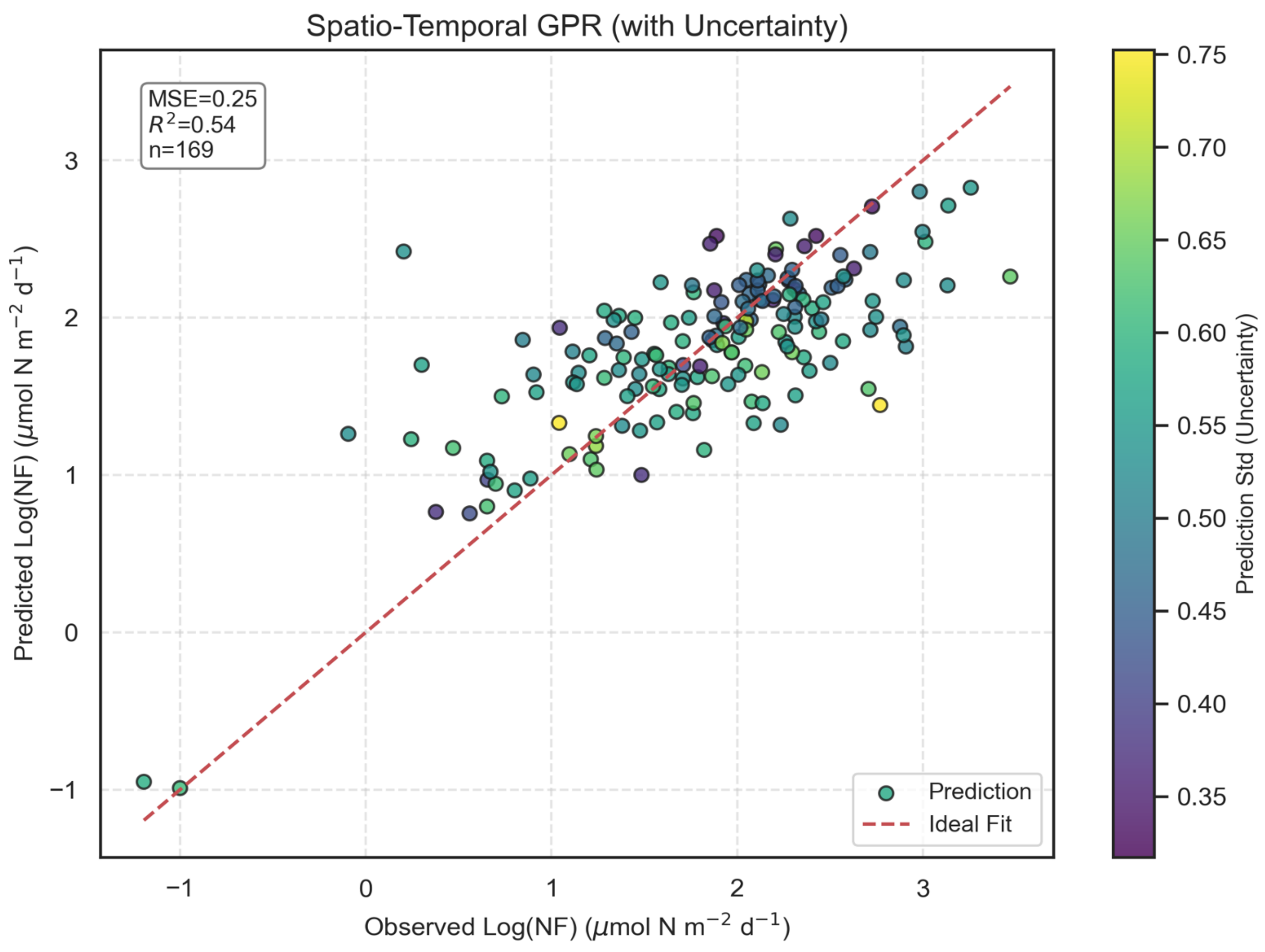
- Modeled:
    - Nonlinear effects of a single environmental variable (MLD, Salinity, DO, etc.).
    - Important interactions among core environment variables (such as CHL and N/P/Fe).
    - Joint smoothing of space (coord1, coord2) (i.e., geographical distribution).
    - Time (time_sin, time_cos) is used as a periodic signal to model seasonal variations.
- Partial dependence plots (PDPs) to show thresholds/turning points.

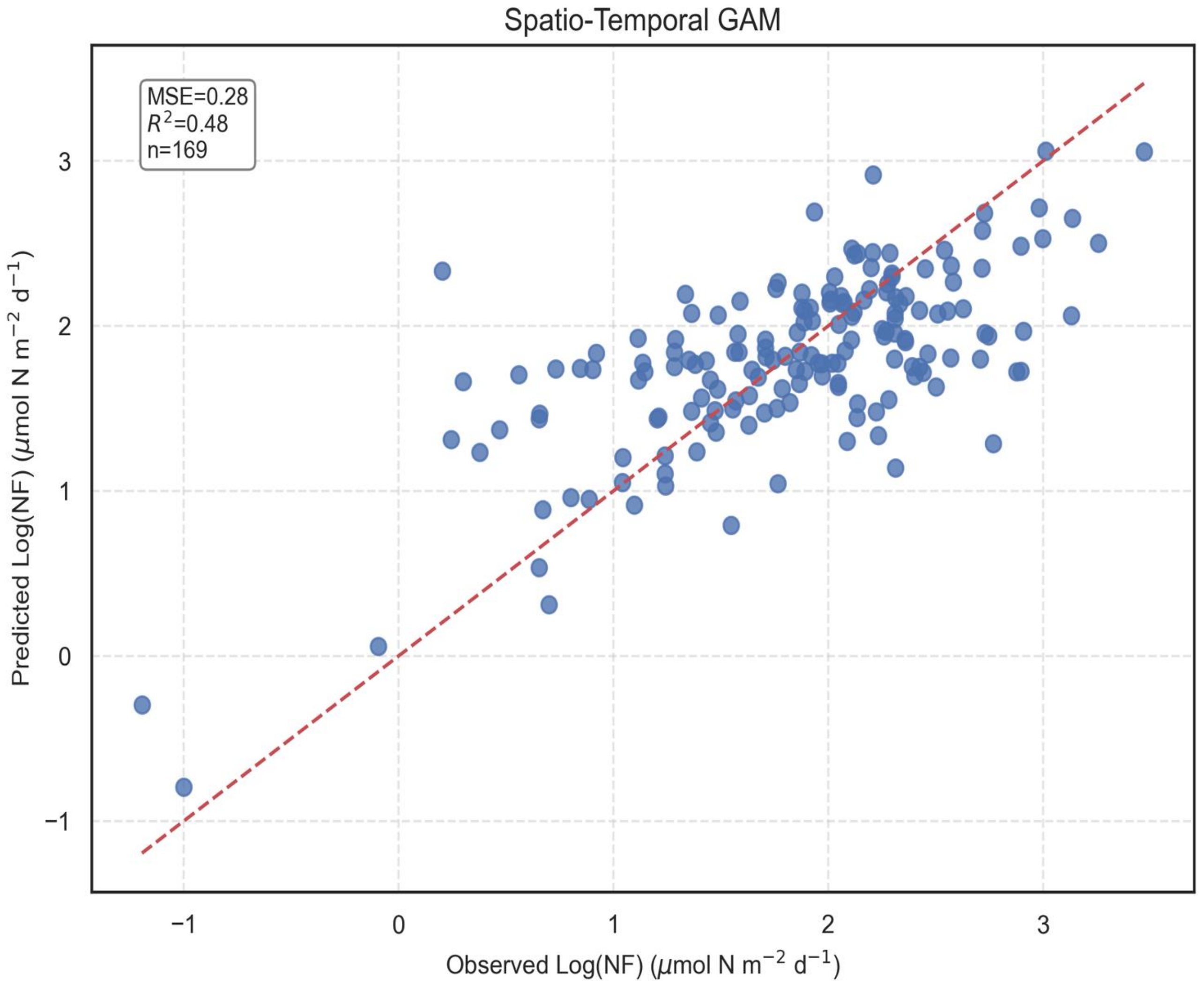
Evaluation:

- Metrics: R^2, MSE (compare all models).
- Visualization: Prediction maps (all models), PDPs (GAM), uncertainty (GPR), regional insights (GPR).

Results & Findings





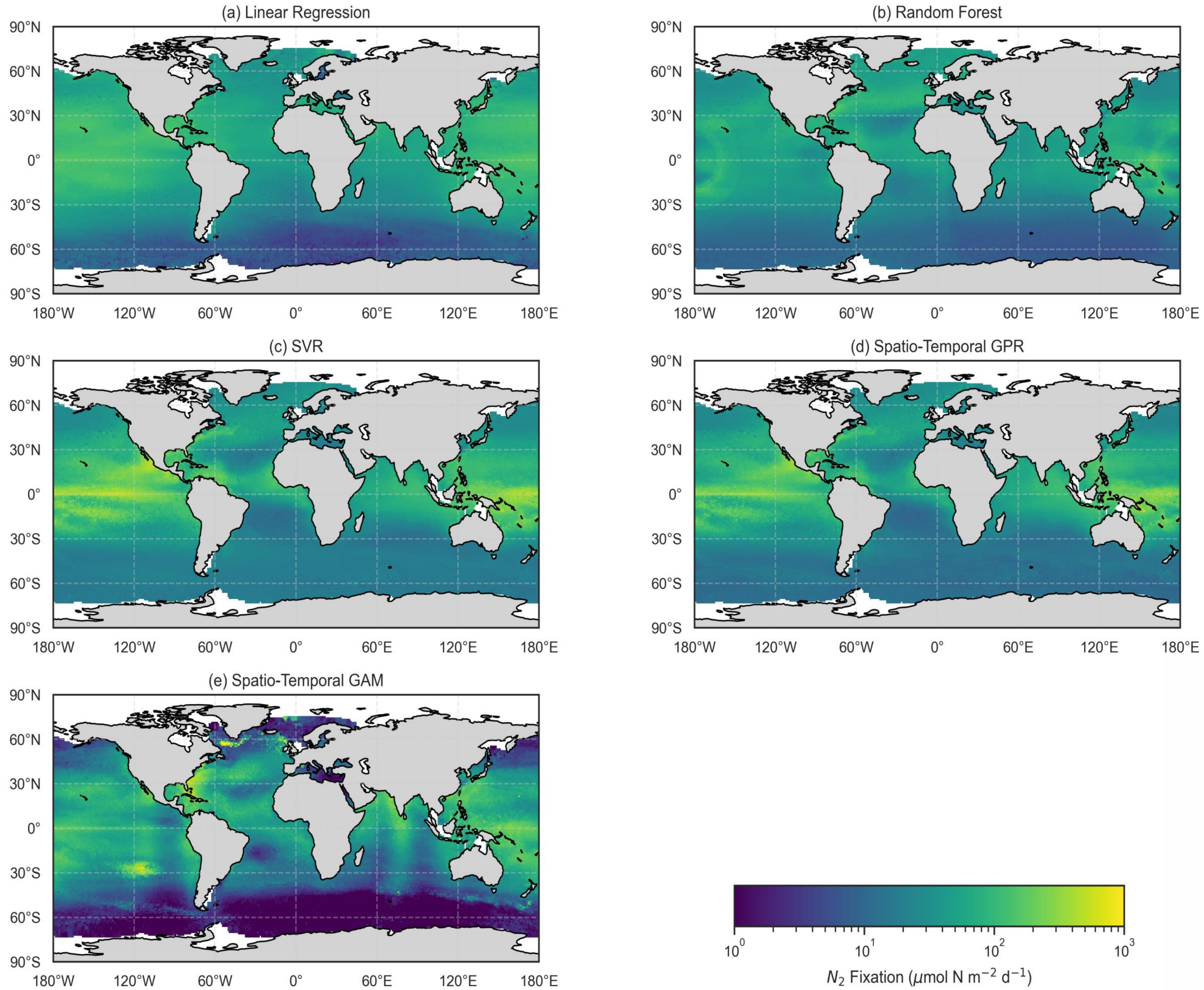


Model	R^2	MSE
LR	0.15	0.46
RF	0.37	0.34
SVR	0.37	0.35
GPR	**0.54**	**0.25**
GAM	**0.48**	**0.28**

Model Performance

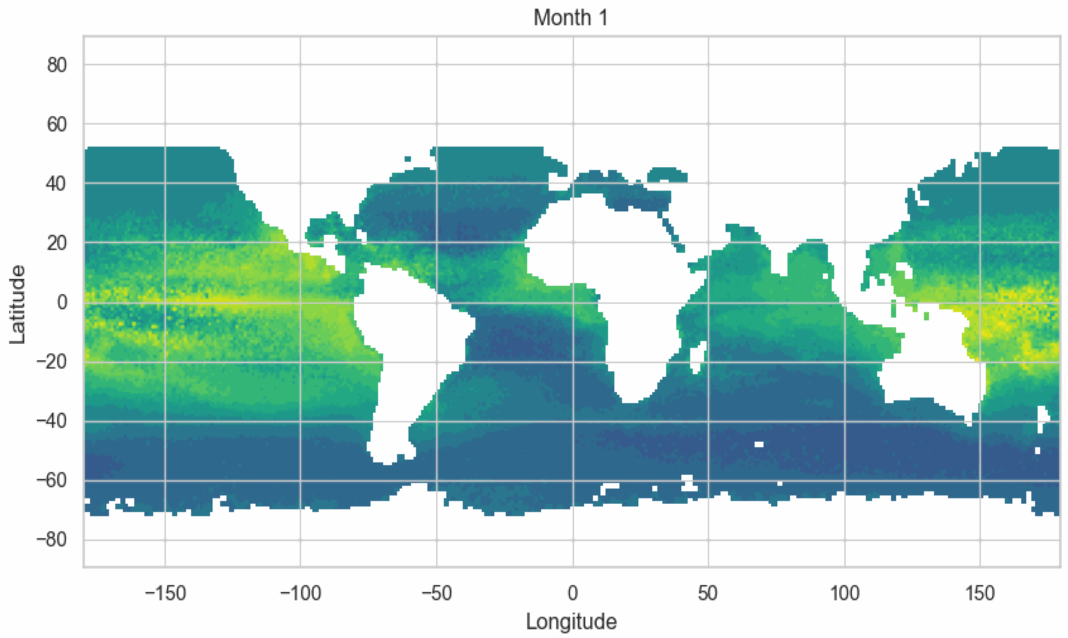
- Spatio-Temporal GPR/GAM outperformed ML models (LR/RF/SVR) in R^2/MSE.
- GPR quantifies uncertainties.

Results & Findings

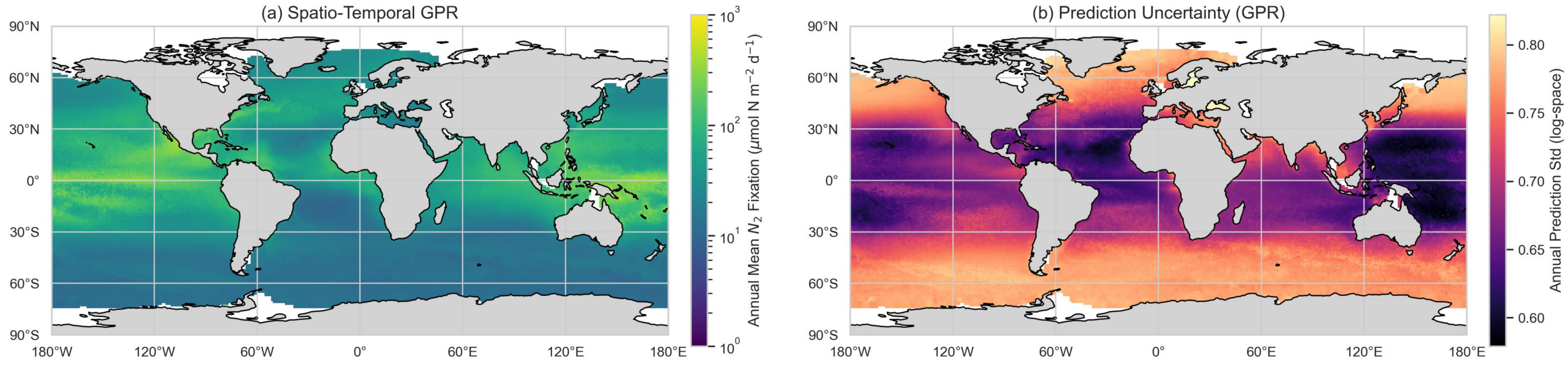


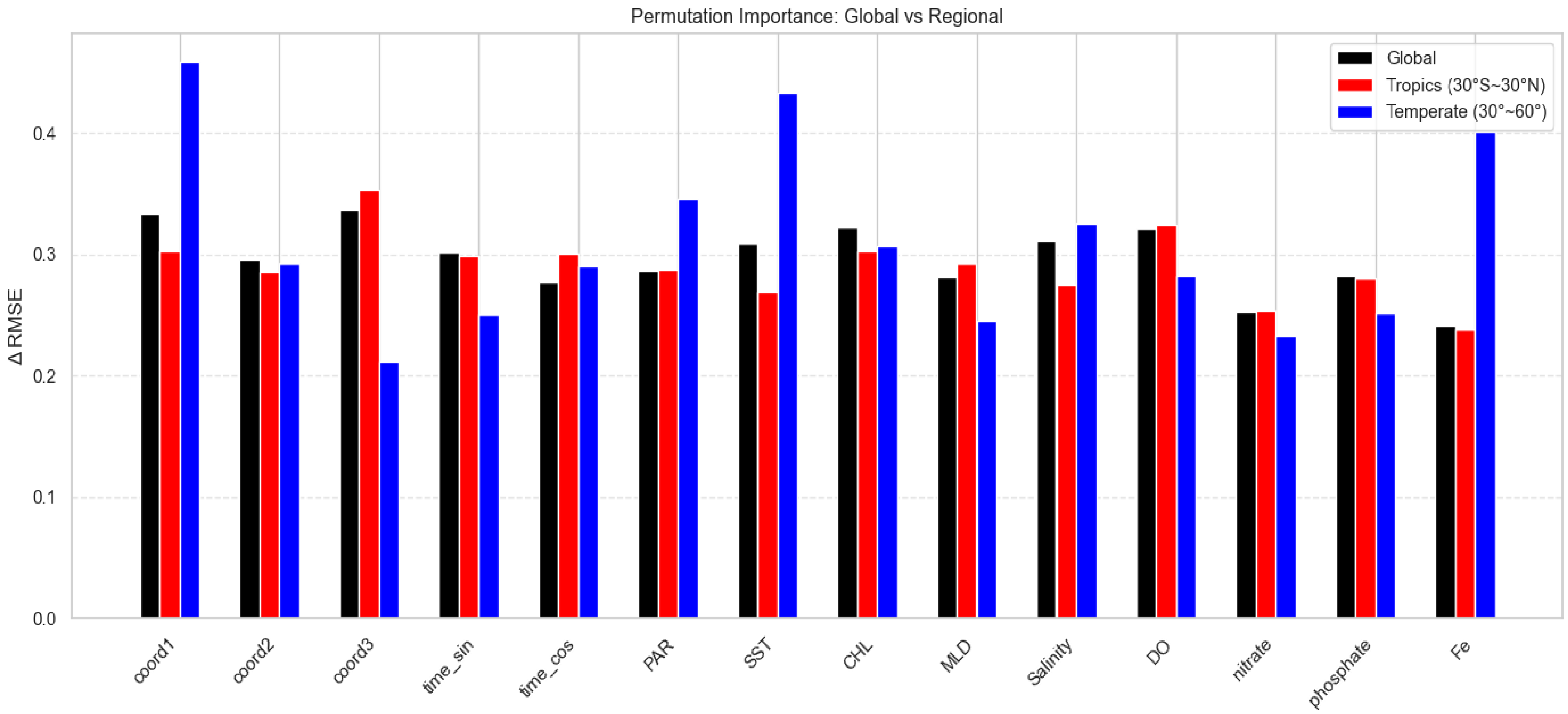
Global Predictions:

- Models agree on some high/low fixation zones (e.g., subtropical gyres, upwelling regions, South Ocean).
- RF/SVR/LR: May oversmooth spatial heterogeneity (no explicit spatial kernels).
- GPR/GAM: Likely capture finer-scale variations (e.g., sharper transitions at ocean fronts - Gulf Stream Front, Equatorial front).



Results & Findings



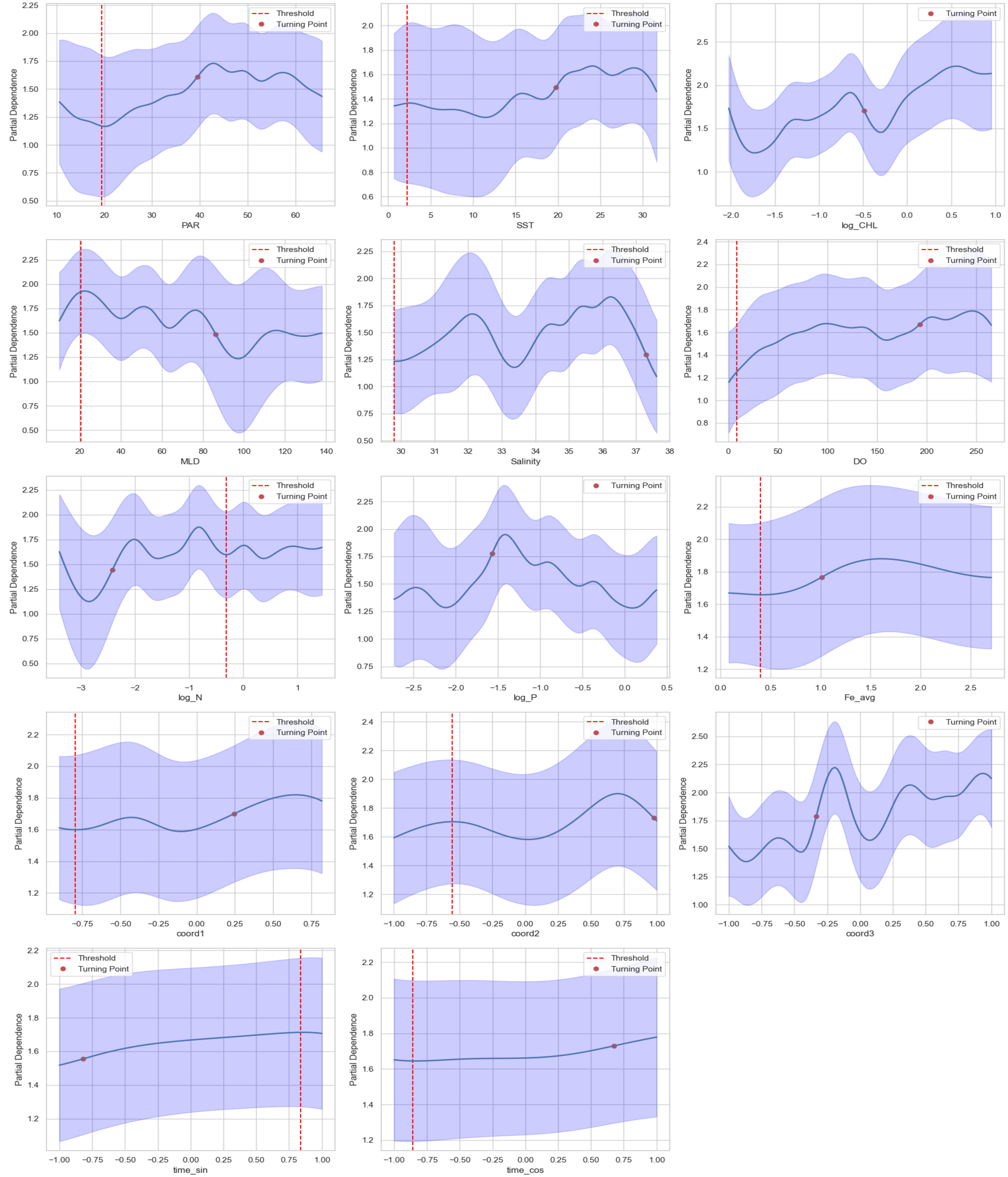


Uncertainty Patterns

- High uncertainty areas: Data-sparse regions (e.g., Southern Ocean) or dynamic zones (e.g., equatorial upwelling).
- Low uncertainty: Where training data is dense (e.g., North Atlantic, Pacific).

Permutation Importance

- **Global**: Dominant drivers (e.g., SST, CHL, DO) **align with Tang et al. (2019)**.
- Tropics: Expect DO/CHL/SST as top predictors (warm waters, light).
- Temperate: SST/Fe/PAR/Salinity dominate (light/mixing-driven fixation).



Results & Findings

PDP	Blue line	influence of variable changes on the prediction results
Threshold	Red line	saturation point
Turning Point	Red point	The position with the most sensitive response changes

Environmental Variables

- SST: Sensitive ~20°C; PAR: Threshold ~20 (fixation peaks in warm waters, declines beyond due to thermal stress)
- Fe: Saturating curve (fixation plateaus above ~0.5 nM, suggesting iron sufficiency).

Spatio-temporal Variables

- time_sin: Sensitive in summer months, saturation point in winter.

  `time_sin ∈ (0.75, 1]` `month ∈ (7.62, 9)` -

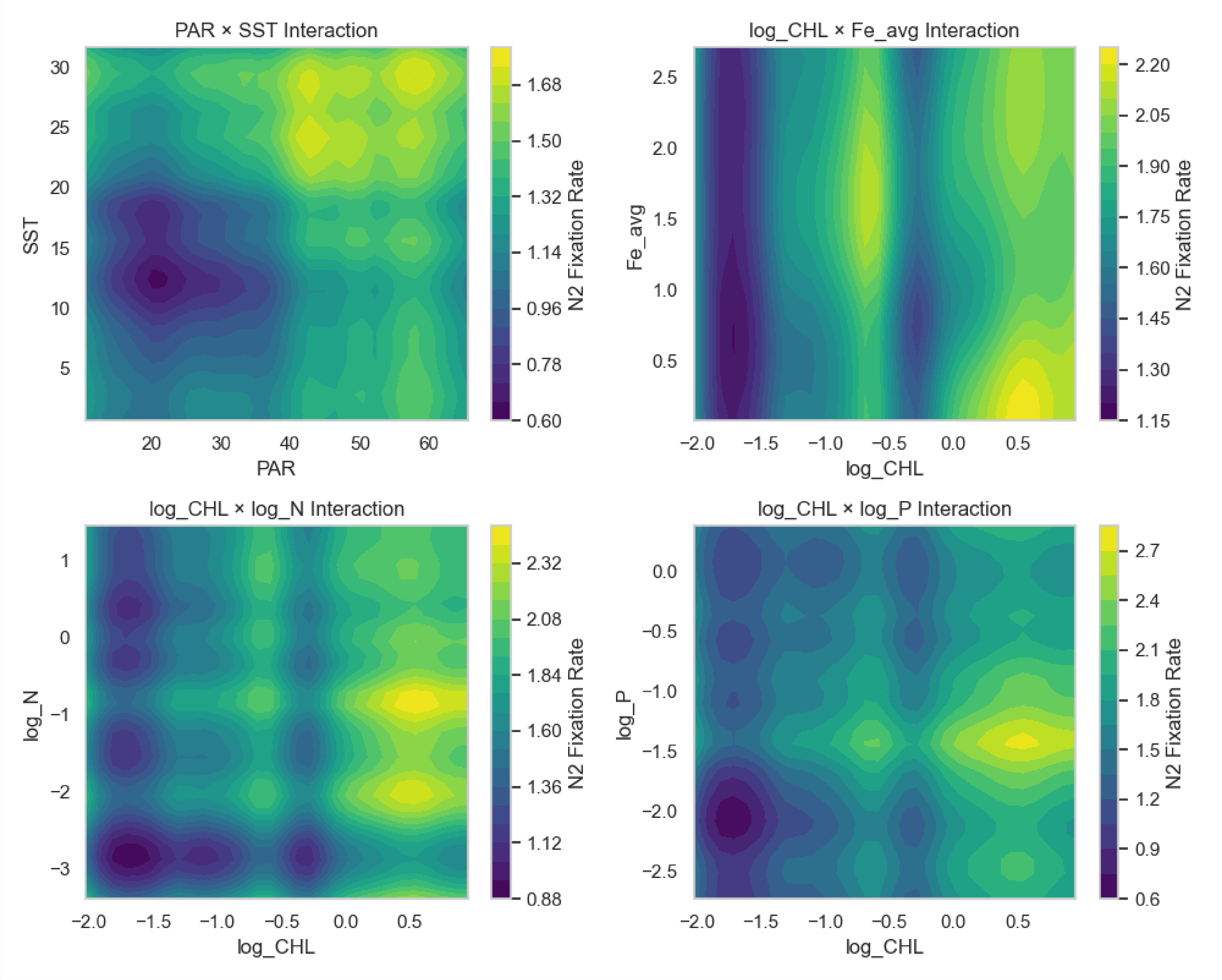
  `time_sin ∈ [-1, -0.75)` `month ∈ (10.38, 12)`

- Coord1: Latitude-dependent thresholds (sensitive in ~15°N, saturation point in ~48°S).

  `sin(lat) = -0.75` `lat = arcsin(-0.75) ≈ -48.59`

  `sin(lat) = 0.25` `lat = arcsin(0.25) ≈ 14.48°`

Results & Findings



Environmental Variables Interactions

- (1) **High PAR + SST 25~30°C** maximizes fixation
- (2) **Iron restriction remission** may occur (the CHL effect is weak at low Fe and enhanced at high Fe).
- (3) **The negative effect of CHL**: The color of high CHL area (~ -0.5) change darker - the competition among phytoplankton (non-nitrogen-fixing organisms consuming resources).
- (4) **Nitrogen competition signal**: the CHL effect weakens at high N.
- (5) **Log[P] ~ -1.5 + High CHL** maximizes fixation. High P: The negative effect of CHL is enhanced. → Phytoplankton competition dominates when **P is sufficient**.

Discussion & Implications

4.1. Environmental Controls on Marine N_2 Fixation

In the study of Luo et al. (2014), DO_{min}, I_s, and SST are found to be the best predictors of N_2 fixation at the global scale. DO_{min} might affect the extent of nitrogen loss in the water column leading to conditions favorable for diazotrophy with an excess of phosphorus ($P^* > 0$). Meanwhile, solar radiation has been hypothesized to constrain the biogeography of diazotrophs in light of their large energy requirement (Mahaffey et al., 2005). In contrast to Luo et al. (2014), we find that none of the properties studied shows a strong correlation with N_2 fixation rates ($R^2 < 0.1$ for all the predictors in Table 1 and Figure 2). This is the case even when applying the MLR approach of Luo et al. (2014). We attribute the lower predictive power to the additional observations in our meta-analysis further confounding the relation of N_2 fixation to predictors. For example, low N_2 fixation rates were recently observed in the eastern South Pacific despite the low DO_{min} and high P^* (Knapp et al., 2016). In addition, organic matter as a source of energy in heterotrophic N_2 fixers may partially offset the light energy requirement (Rahav et al., 2016), thereby broadening the potential niche of N_2 fixation and weakening the correlation to surface solar radiation or to SST. This is in line with the recent discovery of diazotrophs in colder waters (Blais et al., 2012; Fernández-Méndez et al., 2016). We also did not find significant correlations of N_2 fixation rates to modeled Fe:N and P:N nutrient supply ratios or subsurface nutrient concentrations (Figure S6 in the supporting information). Overall, these results challenge some of the traditional assumptions of what regulates N_2 fixation in the ocean. It also implies that simple regression analyses may not capture the complex interplay of environmental factors regulating the distribution and magnitude of N_2 fixation.

Tang et al. 's Contributions

- No single predictor strongly correlated with N_2 fixation - demonstrated the complexity of N_2 fixation regulation.
- Highlighted discrepancies between models.

My project

Contributions:

- Add dissolved iron (Fe) as a predictor and use 2023 World Ocean Database to address gaps.
- Demonstrated spatio-temporal GPR's superiority over traditional ML in capturing uncertainties.
- GAM's interpretability revealed actionable thresholds and interactions between variables (e.g., for nutrient management).

Limitations:

- GAM's spline-based interactions may oversimplify extreme conditions.
- Due to time constraints, no detailed temporal analysis was conducted in GPR because the influence of time is less than that of space.

Reflection & Takeaways

Pros/Cons:

- **GPR**: Ideal for uncertainty-aware applications; accurate but computationally heavy.
- **GAM**: Reveals mechanisms (non-linearities); Interpretable for interactions between space, time and environmental features, but less flexible for abrupt changes.
- Traditional ML is simpler but less nuanced.

Future Work:

- **Hybrid models** (e.g., Spatio-temporal GAM + spatial random effects).
- Include **more temporal granularity** (e.g., daily data) to conduct temporal analysis.

Reference

- Luo, Y. W., Lima, I. D., Karl, D. M., Deutsch, C. A., & Doney, S. C. (2014). Data-based assessment of environmental controls on global marine nitrogen fixation. *Biogeosciences, 11(3)*, 691-708. https://doi.org/10.5194/bg-11-691-2014.
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- Varga, B. (2022). Gaussian process-based spatio-temporal predictor. *Acta Polytechnica Hungarica, 19(5)*, 69-84.
- Wang, W. L., Moore, J. K., Martiny, A. C., & Primeau, F. W. (2019). Convergent estimates of marine nitrogen fixation. *Nature, 566(7743),* 205-211. https://doi.org/10.1038/s41586-019-0911-2.

THANK YOU!